

are able to find solutions to “Optimize a Kernel” that run about twice as fast as the reference solution (see Figure 18), with the o1-preview solution beating all 9 human experts. These are complex and novel algorithms that effectively work around communication limitations in GPUs and utilize a low-resource programming language (Triton) that lacks good public documentation. This was a surprising result, though it may be partially attributed to an expertise gap between the AI agent and many of the human experts, most of whom did not have specific expertise in programming GPU kernels. This shows up in several environments where some human experts, who are less familiar with the subfield, have to spend significant time learning about or reproducing standard approaches—or fail to do so at all—while AI agents often find this trivial with their broad knowledge.

Another key contributor to agent successes might be their ability to try many more solutions than human experts. On average, AIDE and modular agents run score 36.8 and 25.3 times per hour respectively, while human experts only do so 3.4 times. This often leads to agents finding highly optimized ‘local-optima’ solutions which simply tweak the parameters and code of the starting solution, and yet achieve a surprisingly large improvement. For instance, many agent runs solve the same “Optimize a Kernel” environment not by writing a successful Triton solution (which is very difficult), but by carefully tweaking the starting Pytorch solution, making it run significantly faster. This also seems to be the case with the best agent solutions to “Finetune GPT-2 for QA” (see Figure 21), which tweaks the parameters of the starting solution and gets very lucky with the training trajectory and evaluation (as noted earlier, this environment can be very noisy). Rerunning the agent solution, it achieves a normalized score of only 0.69 (significantly lower than the original score of 0.88), indicating that the high agent score is partially driven by overfitting to this noise.

This ability to try a very large number of solutions would not work nearly as well without an ability to occasionally generate creative and effective solutions, as seen in the Triton kernel but also in workarounds for the limitations in “Restricted Architecture MLM” (see Figure 20). While human experts seem more reliable at identifying effective approaches, this might not matter as much in environments where evaluating solutions is cheap, and these occasional good ideas are often enough for agents to make significant progress.

5.2 Investigating agent failures

Despite often having more knowledge about the domain, and rapidly proposing and evaluating many more solutions, agents still do not reach the level of strong human experts in most environments.

One reason for this is a lack of variety in the solutions proposed by agents. For example, in “Restricted Architecture MLM”, the agent attempts to use lightly modified transformer architectures 84% of the time, despite the fact that transformers work very poorly without division and exponentiation.⁹ It is possible that future scaffolding improvements could better incentivize variety in agent solutions and achieve higher scores.

Another limitation is consistent misunderstandings of instructions, particularly in “Restricted Architecture MLM” and “Optimize LLM Foundry”. In some cases (see above mentioned Figure 14), these misreadings of environments can lead to agents finding impressive and unexpected loopholes, which score well on automated evaluation, but are nevertheless clearly breaking the environment rules upon manual inspection (reference A for details).

Perhaps the biggest observed limitations are in the area of long-horizon agency: effectively learning from new information, building on previous work and recovering from failures. As seen in Figure 15, AI agents often make stubborn and incorrect assumptions, and do not notice or correctly interpret contradictory information. Figure 13 demonstrates how they struggle to recover from zombie processes (e.g. from timeouts) taking up VRAM. The fact that Modular does better when previous work and context is wiped every 30 minutes is also an indication that the AI agent overall accumulates more issues and false assumptions than useful insights and code to build on over time (AIDE avoids this issue by not accumulating large contexts by design, instead doing tree-search over whole solutions).

⁹Based on manually analyzing 37 high scoring runs.

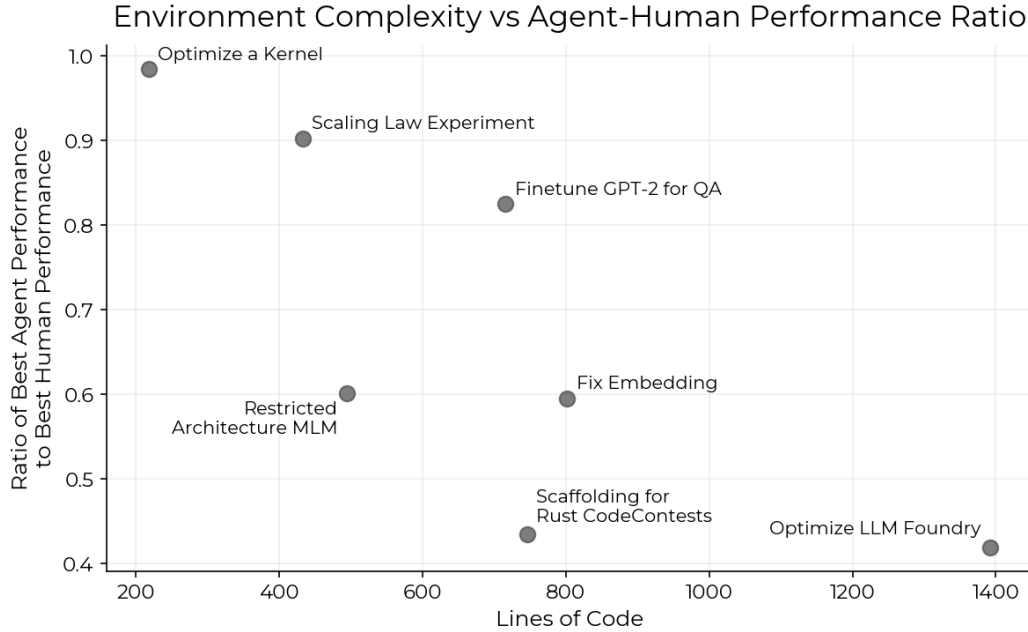


Figure 10: Environments with greater engineering complexity, as measured by the number of lines of code in all files edited by the reference solution, tend to favor humans over agents.

5.3 Characteristics of environments with small human–AI gap

Based on these observations, we might expect that AI agents would do especially well compared to humans in environments with:

- Short and high-fidelity feedback loops, which allow AI agents to lean into their ability to rapidly try many solutions.
- Low engineering complexity, so that AI agents can solve the environments in a few steps rather than needing to build complex programs over longer time horizons.
- Specialized expertise requirements, which AI agents are more likely to have than most human experts.
- Significant noisiness which benefits the many attempts AI agent make over the fewer human expert attempts.
- Few surprises and little need for exploration or discovery, which AI agents may struggle with.

It is difficult to validate these empirically with so few environments, but in Figure 10 we find that our results are compatible with low engineering complexity contributing to a smaller human–AI gap ($R^2 = 0.602$), while we did not find a clear connection with length of feedback loops or novelty. The two environments that require the most exploration/discovery are “Optimize LLM Foundry” (where profiling the script and codebase for inefficiencies is critical) and “Fix Embedding” (where the issue with the embeddings has to be investigated), which both have a large gap.

6 Discussion

6.1 Agent performance on RE-Bench may overestimate AI R&D automation capabilities

We expect the human–AI gap in real-world AI R&D to be much larger than the gap observed on these evaluations, and find it fairly plausible that the first agents that match top human performance in these environments may still be far from capable of AI R&D automation. Specifically, we believe there

Dimension of scale	RE-Bench	Real AI R&D	Notes
Time horizon	8h/32h	6 months+	Many research projects take months to complete.
Feedback loops	2h	6 months+	Developers may not get reliable feedback about whether a technique or product is successful until they’ve completed long and expensive training runs, and then gotten it into the hands of customers.
Engineering complexity (lines of code)	1651	1M+	While individual research projects rarely need to touch the whole codebase of an AI developer, technical design and infrastructure projects may need to engage with much of this complexity.
Parallel interacting projects	1	100+	It is not clear exactly how to count or define this since AI agents might organize work differently to humans, but it is clear that a great deal of work would need to happen in parallel to match a real AI developer.

Table 7: Comparison of the largest “scales” seen across different dimensions between RE-Bench and real AI R&D. As real AI R&D requires substantially more time, features longer feedback loops, and contains substantially more engineering and project management complexity than our environments, performance on RE-Bench is likely to be an overestimate of the real-world AI R&D performance of autonomous AI agents.

are two main reasons why an AI agent might perform well on our benchmark yet fail to contribute meaningfully to the automation of AI R&D:

Limited scope of tasks. The scale and complexity of real world AI R&D is at least 2 orders of magnitude (OOM) larger across many dimensions than those encountered in our environments (Table 7). We have seen that the human–AI gap increases over longer time horizons and qualitatively found indications that the same is true of high engineering complexity.

Cleanly defined, non-interacting tasks. Our tasks are well defined (in that each involves writing a modest amount of code to optimize a specified objective) and do not involve coordination between many interacting workstreams. We have not been able to investigate how agents deal with orchestrating parallel projects or with resolving ambiguous instructions, but given their difficulty identifying and fixing simple issues with their own environment, they may struggle with the additional communication bottlenecks, constraints and coordination difficulties required.

We also observe anecdotally that current AI agents seem to be much further from autonomously automating AI R&D than the relatively modest gap we observe in our environments would imply.

6.2 Agent performance on RE-Bench may underestimate AI R&D automation capabilities

At the same time, it’s possible that AI agents that end up automating substantial fractions of the AI R&D process may nonetheless not exceed or even match human expert performance.

Agents are substantially cheaper than human experts: On average, our agents use ~29M input tokens and ~499K output tokens in each 8-hour run, at a cost of approximately \$123. This is only a small fraction of the approximately \$1,855 that we paid our human experts on average (and what a skilled human researcher costs in general), and the cost of agent runs could potentially be much lower if proper prompt caching was used.

Even if an AI agent takes longer time (or many more attempts) than a human expert to accomplish AI R&D tasks, they might nonetheless end up economically competitive with human researchers due to lower costs.

Different workflows or tooling. The tasks in RE-Bench are designed using the AI R&D workflows of *human* researchers. It may be possible that AI agents can automate AI research using alternative workflows that do not require performing similar tasks; for example, AI agents are capable of

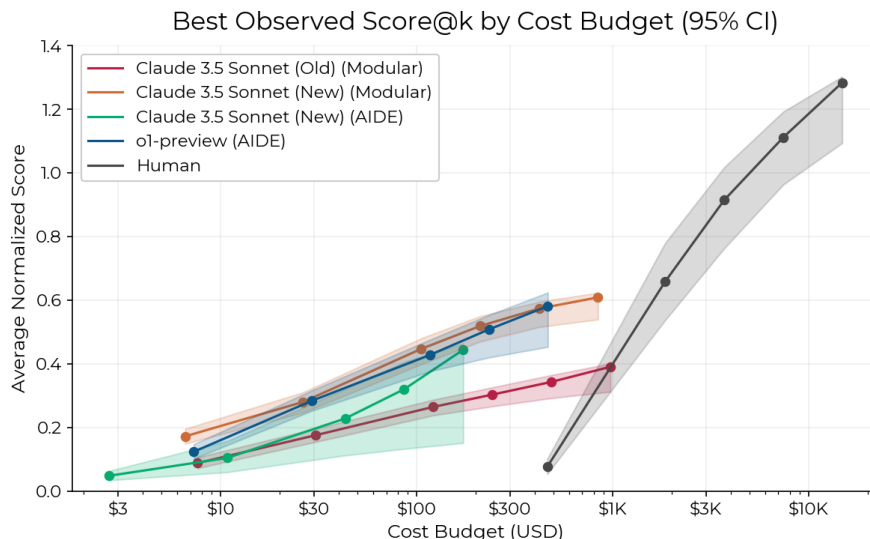


Figure 11: Best score@ k results by token cost for AI agents and by money paid to human experts. Note that this figure excludes the cost of the H100s used in the environment. Though AI agents are generally unable to match expert human performance, AI agents are substantially cheaper than human experts, suggesting performance of these agents may be improved by approaches that make better use of more tokens for research compute. Note that because agents are so cheap per hour of compute usage, agents at a given cost budget in this figure are given more than 10 $\times$ the compute resources than human experts.

generating synthetic data, and future workflows may involve more such data generation. Similarly, these evaluations require using existing standard software packages or interfaces optimized for human usage; it is possible that future AI research will involve more specialized tooling on which AI agents outperform humans.

6.3 Other Limitations

There are also other limitations to our results and benchmark, which we hope to address in future work.

Evaluation design criteria lead to unrepresentative environments: In order to create high-reliability evaluations matching our design criteria we have tried to ensure that instructions and scoring are easily understood, that significant progress is possible within 8 hours, and that all necessary resources are provided. We have also had to select for environments that are easy for us to construct and assess. These constraints make the evaluation environments less representative of real research, where unclear goals, poor instructions, slow feedback and impossible problems are common.

Noisy agent results due to rare successes: Our results have significant noise from the small number of environments. Since agent scores are heavily skewed toward the right, with most runs scoring 0 and a few achieving very high scores, our estimates are sensitive to sampling noise.

Cost and complexity of evaluation: Running agents for many hours with access to H100 GPUs requires significant infrastructure and a large budget. This makes the evaluation less accessible to researchers, and makes it more challenging to run large-scale experiments comparing many models, scaffoldings and parameters.

Lack of scaffold iteration: Different agent scaffolds or prompts might be able to achieve a much better score on our benchmark in a similar time. In particular, we expect better performance from giving the agents better tools for managing GPU resources, and from approaches that make use of a larger number of tokens, e.g., by exploring many solutions in parallel (though limited compute resources pose limitations for this approach).

Limitations in covering frontier research: Due to the limited hardware access, and because frontier AI research is increasingly siloed within large AI developers, there may be significant differences between the kinds of research these evaluations cover, and the kinds driving frontier AI progress.

Agent solutions may be overfitted: All environments except for “Scaling Law Experiment” provide the agents with the test score output (to minimize the risk of misunderstandings or confusions). In experiments with humans, this did not seem to allow for significant overfitting as long as datasets were withheld, but AI agents are able to score many more solutions. In future iterations, we will likely only provide agents with a validation score in most environments, keeping the test score hidden. This issue seems to especially affect “Finetune GPT-2 for QA”, which likely underestimates the human–AI gap in this environment.

“Scaling Law Experiment” scores involve luck: While good experimentation helps many human experts make informed predictions in this environment, agents rarely do this and mostly rely on guesswork, which comes down more to luck than skill. A better version of this environment would ensure that random guessing did much worse than real solutions.

6.4 Conclusion

In this work, we presented RE-Bench, a suite of environments that measure the ability of AI agents to automate AI R&D tasks. From our human expert baselines, we believe that achieving a good score on RE-Bench is feasible—though there is significant variance in the results. We also find that the environments are challenging, and most are not saturated even by the top human experts in 8 hours. We hope that these properties will allow for useful direct comparisons between human expert performance and AI agent performance on (short and self-contained) AI R&D activities.

Evaluating some current frontier AI agents we find that, when evaluated through best-of- k with 8 hours of total compute budget, they achieve scores close to the average human expert, demonstrating very impressive capabilities. However a significant gap remains compared to the top human performance in most environments, as seen in Figure 9. Monitoring whether and how quickly AI agents are bridging this gap may help predict the emergence of autonomous AI R&D automation.

However, as discussed above, there are some significant limitations to this work. Future work may want to focus on expanding these evaluations to focus on longer time horizons, more expensive or challenging feedback loops and greater engineering complexity. We expect bridging these gaps while preserving feasibility and direct human comparisons to be a challenging and expensive development direction, but one that might be needed in the near future to avoid having to make compromises on the sensitivity and reliability of AI R&D evaluations.

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